

Manufacturing Trusted Hardware

Inspection Protocol Sketch for High-Integrity PCBs

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Key takeaway: We present a sketch printed circuit board (PCB) inspection protocol that assumes a well-resourced attacker who knows the protocol and is actively trying to beat it. AI is the motivating focus. As it grows in economic and strategic importance, the ability to verify the underpinning hardware must keep up.

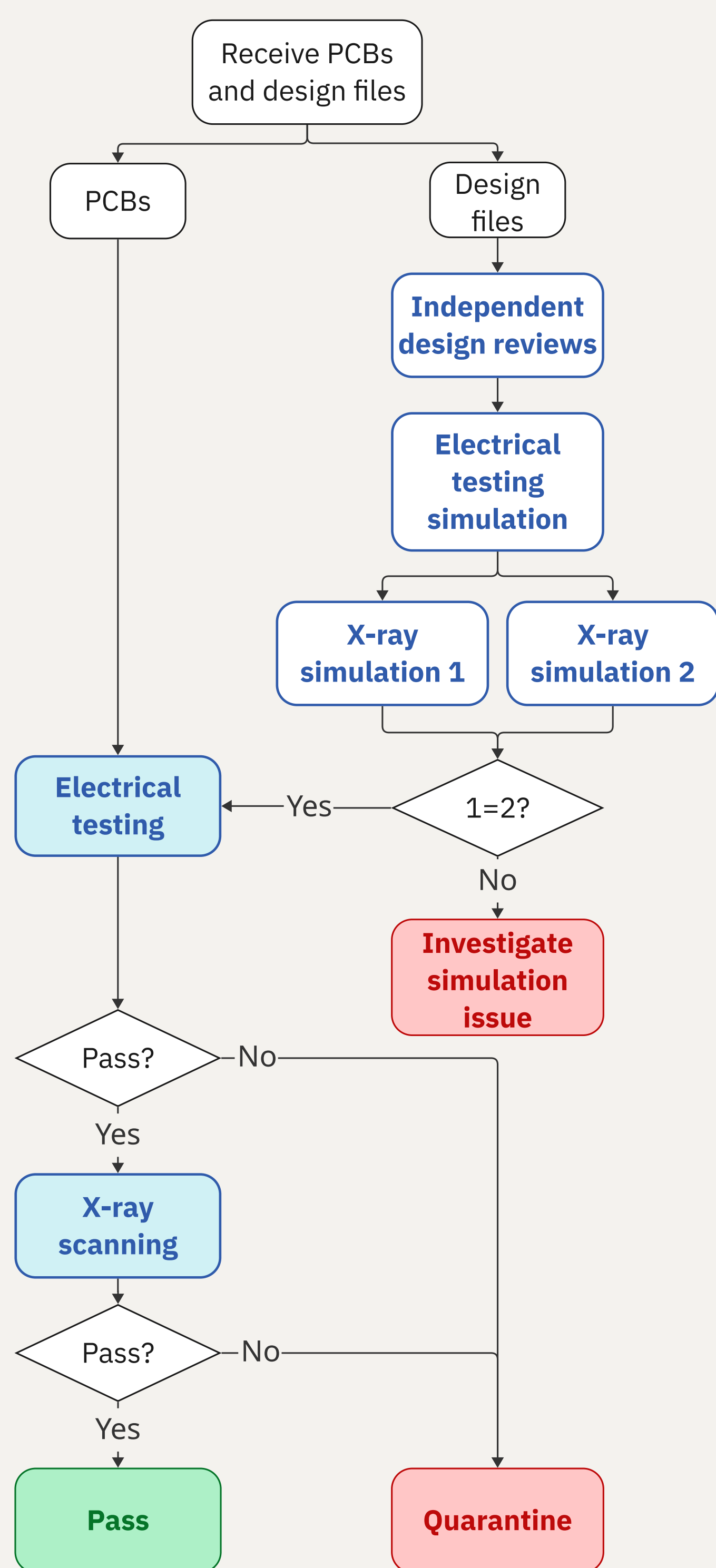
Motivation

The problem: As AI grows in importance, AI hardware becomes a higher-value target. State-level hardware attacks are proven (NSA ANT catalog; Appelbaum et al., *Der Spiegel*, 2013).

The gap: Hardware assurance today is entirely delivered through firmware-level defences which can't detect physical modifications such as implants or trace alterations. No *systematic physical inspection of AI hardware* is known to take place.

This project's scope: Physically inspecting printed circuit boards (PCBs). PCBs are found in almost all compute and networking hardware in AI data centres, and most technical verification solutions for international agreements will rely on them.

The Protocol

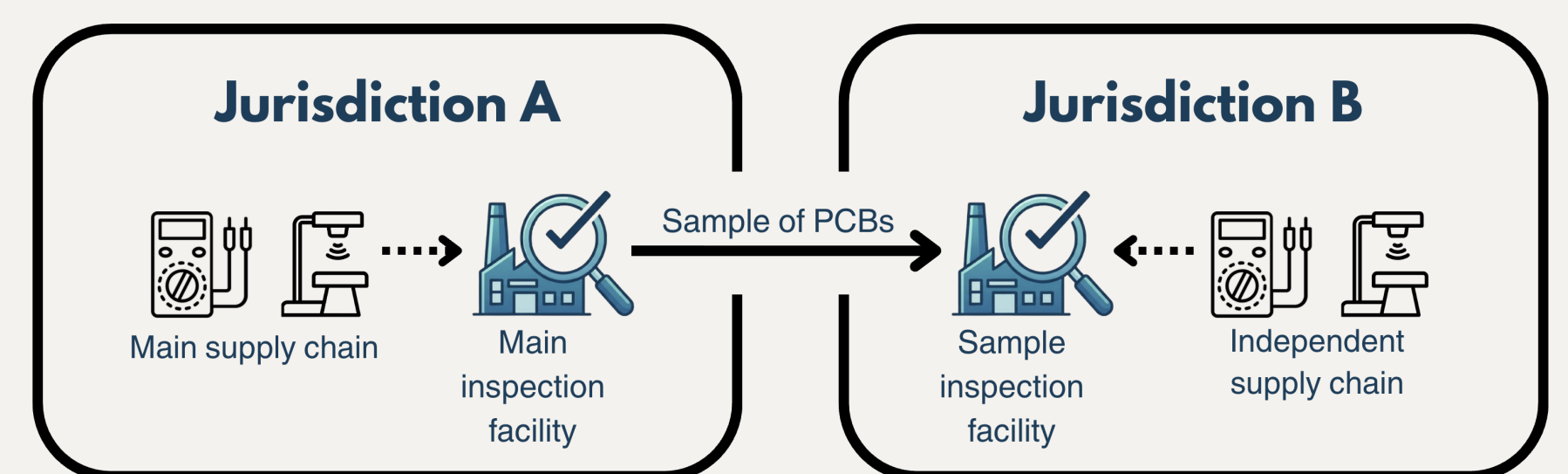


Per board non-destructive inspection flow

Limitations

- **Design file tampering.** If the design files are tampered with and the tampering is not caught, modified boards can pass inspection.
- **Modifications below scanner resolution.** The 3-tiered CT scanning flags suspicious regions. Modifications that evade detection at one level may never receive higher-resolution scans.
- **Compromised inspection facility.** Mitigated by dual-facility inspection in the full protocol; the MVP remains more exposed.
- **Temperature-activated materials.** Some materials become conductive above $\sim 180^{\circ}\text{C}$, evading the protocol's 150°C test.

Dual-facility inspection



Independent verification through geographically separated facilities

Example attack

A state adversary modifies Nvidia H100 GPU server boards during PCB fabrication, adding copper structures that pick up the NVLink signals carrying model activations between GPUs and emit them wirelessly. With H100s in a multi-GPU setup, data over NVLink is *unencrypted* (Gu et al., arXiv:2507.02770, 2025). Intercepting the high-speed signal and reconstructing usable data from it is challenging, but possible for an adversary who controls the board's fabrication.

The server board is installed in the data centre of a leading frontier AI lab. An operative, working in the data centre, places a small receiver (disguised as a sensor) in the rack, collects the data over several days, then retrieves the device — successfully exfiltrating some of the most valuable data in the world. **No known post-fabrication inspection regime would detect this attack.**

Minimum viable protocol (MVP)

- Uses existing commercial computed tomography (CT) comparison software (e.g., Lumafield Voyager), adapted for PCBs, with analyst review.
- Single inspection facility.
- No canary board programme (a red-team exercise: sending in boards with known modifications to verify that the protocol detects them; deferred to full protocol).
- Operational within months with funding.

Next steps

1. **Fund and develop the MVP.**
2. Progress the **full PCB protocol** to defend against a well-resourced nation state by closing R&D gaps. Estimated \$4+M over 3+ years.
3. Build capacity to **increase trust in assembled boards**. Extend the scope to components, integrated circuits, and firmware, possibly requiring spinning up high-assurance manufacturing alongside inspection.
4. **Assure all hardware entering AI data centres.**